IEEE ROBOTICS & AUTOMATION SOCIETY
IUT STUDENT BRANCH CHAPTER

iTest BANGLADESH

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PCB DESIGN CONTEST

ROUND ONE

General Rules

1. Participants must design the PCB according to the given schematic, design requirements, or component list provided in the contest question.
2. Any form of plagiarism or use of pre-made full designs (other than reference libraries) will result in immediate disqualification.
3. The design must be created using the specified EDA tool (**EasyEDA**) as mentioned in the question.
4. **File Submission Requirements:**
   Participants must submit the following files **individually** (*ZIP or compressed file formats are strictly prohibited*):
   - **Main project files:** Both **Schematic** and **Board** files.
   - **PDF layer files:** A **single PDF** containing **all layers** of the PCB. Each page must be **clearly labeled** with the corresponding layer name (e.g. Top Layer, Bottom Layer, Silkscreen, Drill Map etc.).
   - **Report:** A detailed **PCB Design Report** including 2D and 3D views of the final PCB layout.
   - **File naming**: name_uni.filetype
     i.e. Sadi_iut.sch/.pdf/.json

All required files must be submitted before the announced deadline (11.59 AM, 6th October, 2026). Late submissions will not be evaluated.

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Evaluation Criteria

Category	Description	Marks
1. Schematic Design	Proper circuit connections, component labeling, net naming, and clarity of design.	15-20%
2. Custom Library Creation	For creating custom symbols or footprints. Follow given specifications in the question (if any) or datasheet.	5–10%
3. PCB Layout Design	Component placement, routing optimization, trace width selection, power/ground planes, and design compactness.	25-30%
4. Design Rule Check (DRC) Compliance	Must pass all DRC checks. For every logical DRC error, **5 marks** will be deducted.	15-20%
5. Creativity and Design Efficiency	Compact and efficient routing, smart component placement, and thermal considerations.	10-15%
6. Aesthetics and Documentation	Silkscreen clarity, layer management, alignment, proper text placement, and overall presentation.	10-15%

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Additional Notes

- You may need to build libraries according to the given details; if not specified, follow the component's **official datasheet**.
- Designs with **missing connections**, **incorrect net assignments**, or **footprint mismatches** will face point deductions.
- Bonus marks (up to **5 Marks**) may be awarded for **innovative routing** or **excellent PCB design aesthetics**.
- **Alternative versions** of any components, whether through-hole or SMD, may be used in the design if they **meet the given specifications.**

Problem 1: Design a SparkFun Artemis Development Kit (80 Marks)

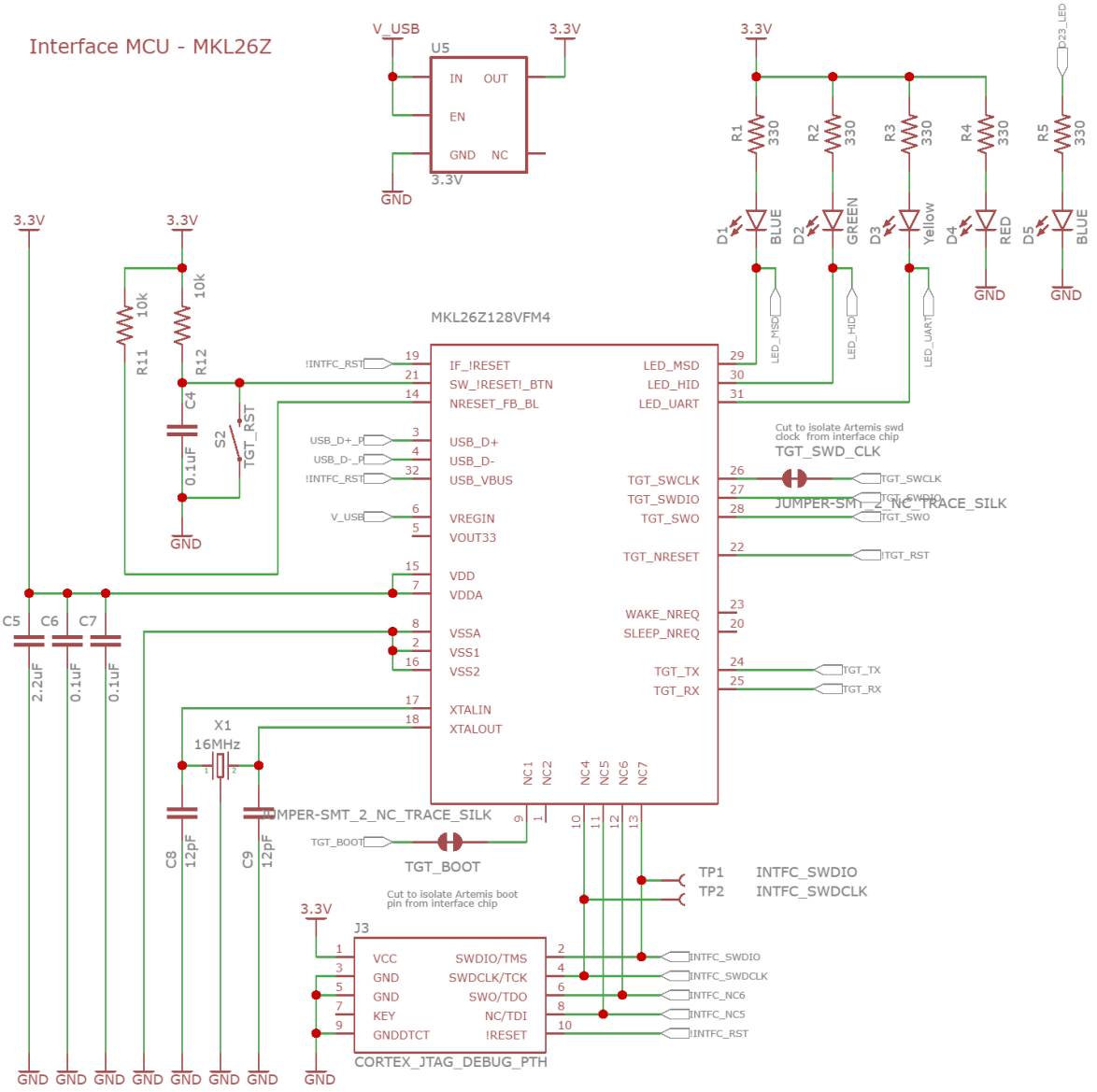
Design the PCB of an embedded development board based on the SparkFun Artemis Development Kit architecture. The board should integrate the Artemis MCU/module, USB/programming interface, basic sensors, camera/interface headers, Qwiic/I2C connectivity, and the required power circuitry.

Design Requirements

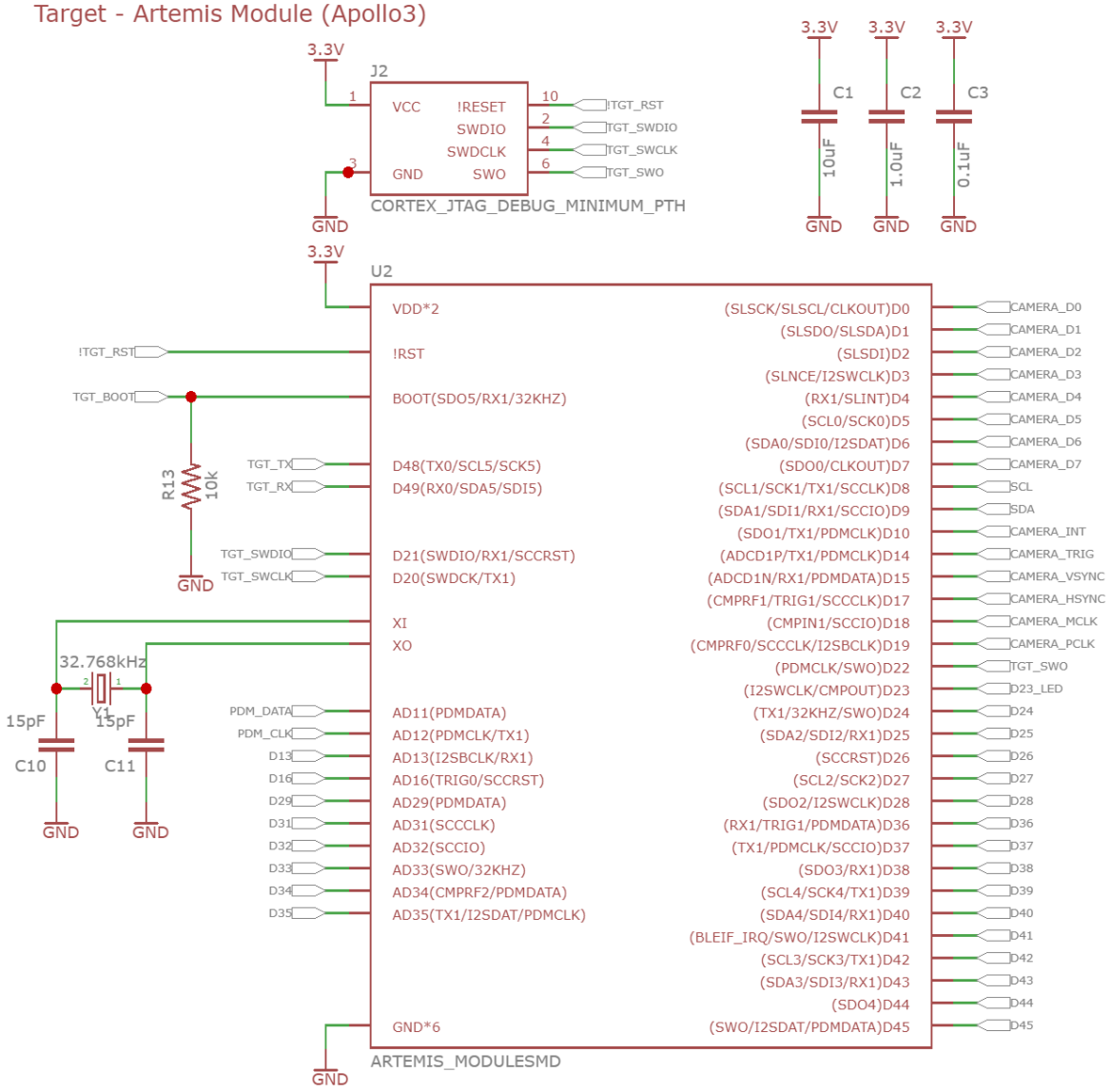
- Use correct footprints and component orientation according to the source schematic and datasheets.
- Place MCU/module decoupling components close to the corresponding supply pins.
- Keep the USB/programming section close to the PCB edge.
- Group sensor and low-level signal circuitry away from noisy power sections.
- Keep I2C/Qwiic routing short and organized.
- Make all external connectors and debug/programming headers accessible.
- Provide clear reference designators and connector labels on silkscreen.
- The finished board must pass the selected EDA tool's DRC.

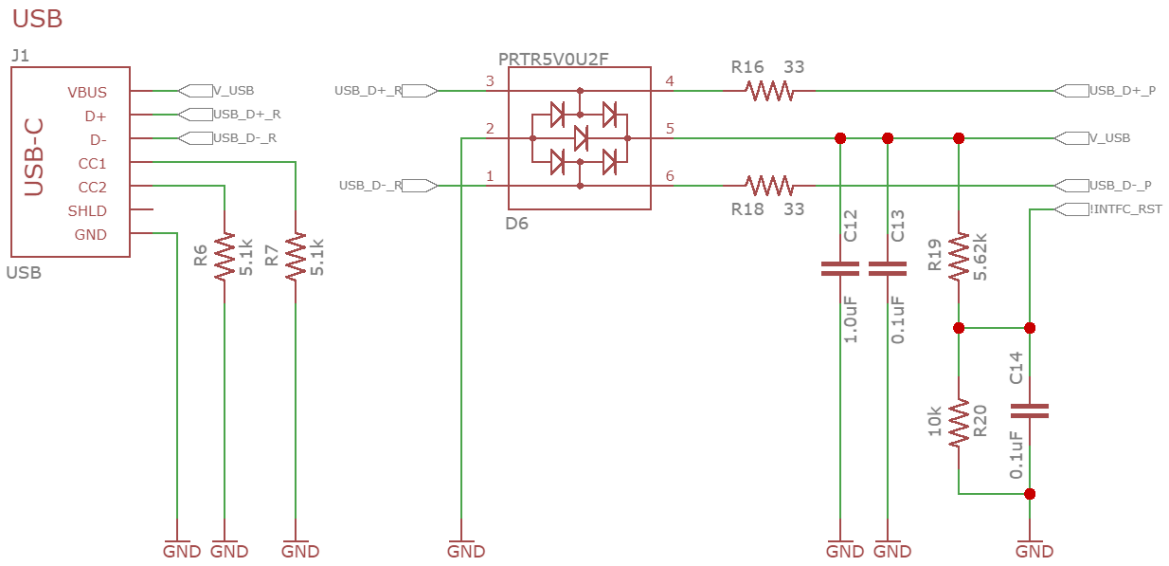
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The Schematic:

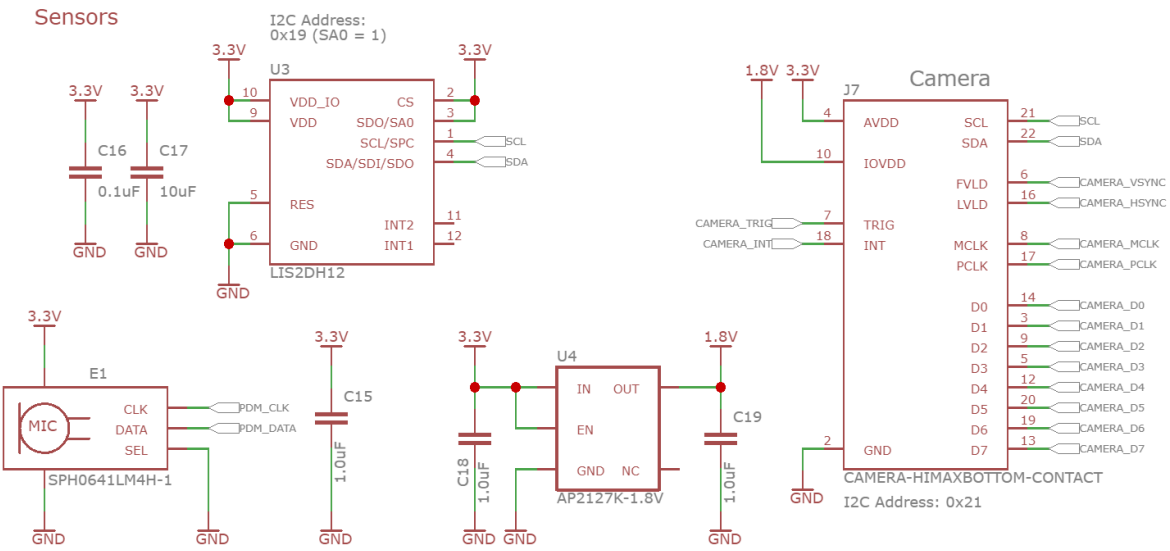


Target - Artemis Module (Apollo3)

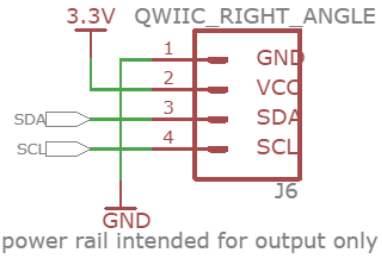




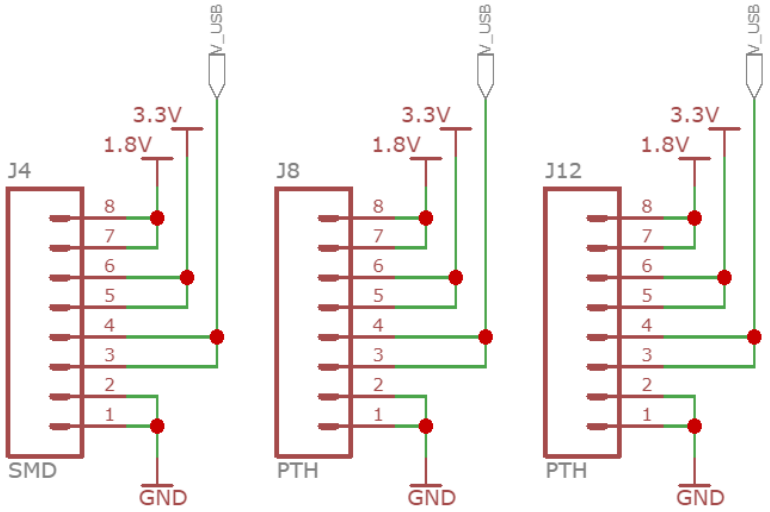
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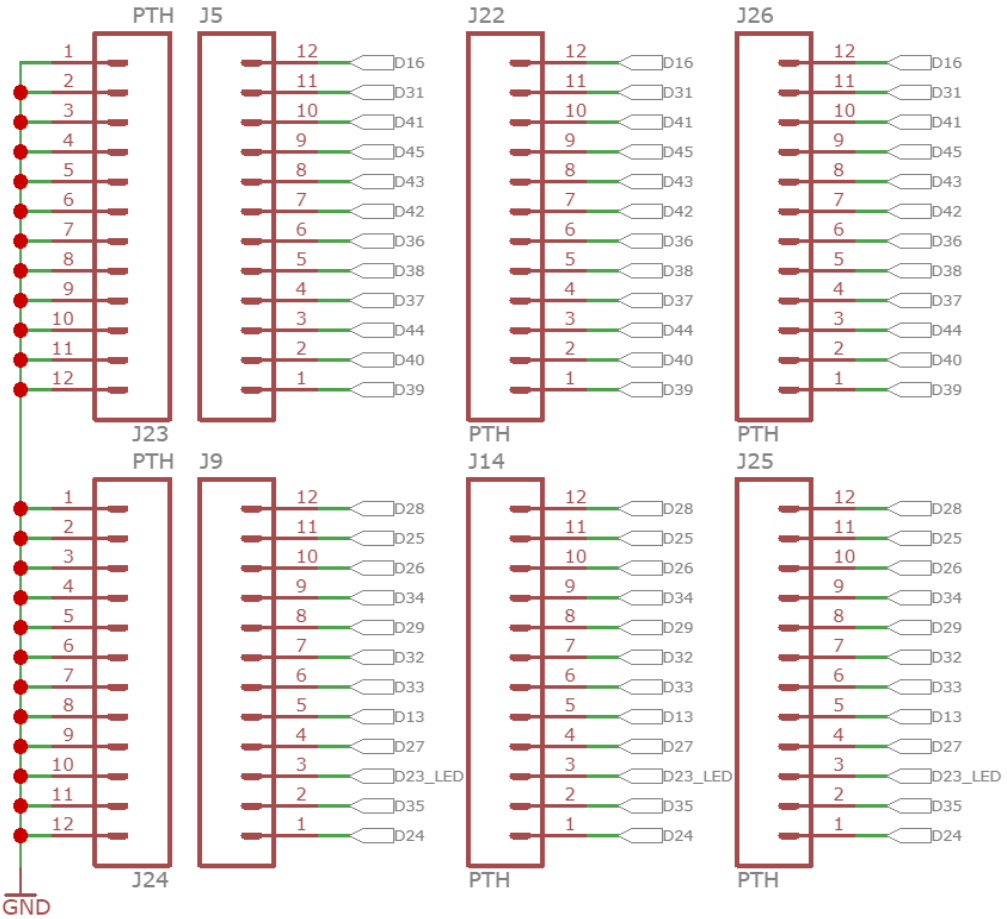


Connectors



power rail intended for output only





Component List:

No.	Component	Designation
1	SparkFun Artemis Module (Apollo3 ARM Cortex-M4F MCU) ARTEMIS_MODULESMD	U2
2	32-bit ARM Cortex-M0+ Microcontroller MKL26Z128VFM4	Interface MCU
3	3-Axis Motion Sensor / Accelerometer LIS2DH12	U3
4	PDM Silicon Digital Microphone SPH0641LM4H-1	E1
5	Low-Dropout (LDO) Voltage Regulator AP2127K (SOT23-5) 1.8V (U4) & 3.3V (U5)	U4, U5
6	Ultra Low Capacitance ESD Protection Diode PRTR5V0U2F	D6
7	Himax CMOS Camera Socket CAMERA-HIMAXBOTTOM-CONTACT	J7
8	USB Type-C Connector USB-C	J1
9	Qwiic I2C Connector QWIIC_RIGHT_ANGLE	J6
10	10-Pin Cortex SWD/JTAG Debug Header	J2, J3
11	12-Pin Expansion Headers	J4, J5, J8, J9, J12, J14, J22-J26
12	32.768kHz Crystal Oscillator	Y1
13	16MHz Crystal Oscillator	X1
14	Tactile Switch / Reset Button	S2
15	Resistor	R1–R20
16	Capacitor	C1–C19
17	Status LEDs (Blue, Green, Yellow, Red)	D1–D5

Datasheets:

- Apollo3 Blue MCU
- MKL26Z128VFM4
- LIS2DH12
- SPH0641LM4H-1
- AP2127
- PRTR5V0U2K
- HM01B0-MNA

Note:

If EasyEDA doesn't have any footprint of any particular component, you may make or download a suitable footprint from the internet. In that case, follow the dimensions given in the datasheets.

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Problem 2: Design a Adafruit Circuit Playground Express(100 points)

Design a compact interactive embedded controller inspired by the Circuit Playground Express. The board should combine a microcontroller, USB interface, sensors, LEDs, buttons, and accessible peripheral connections in a compact, visually balanced form factor

Design Requirements

- Use correct footprints and component orientation according to the source schematic and datasheets.
- Place MCU/module decoupling components close to the corresponding supply pins.
- Keep the USB/programming section close to the PCB edge.
- Group sensor and low-level signal circuitry away from noisy power sections.
- Keep I2C/Qwiic routing short and organized.
- Make all external connectors and debug/programming headers accessible.
- Provide clear reference designators and connector labels on silkscreen.
- The finished board must pass the selected EDA tool's DRC.

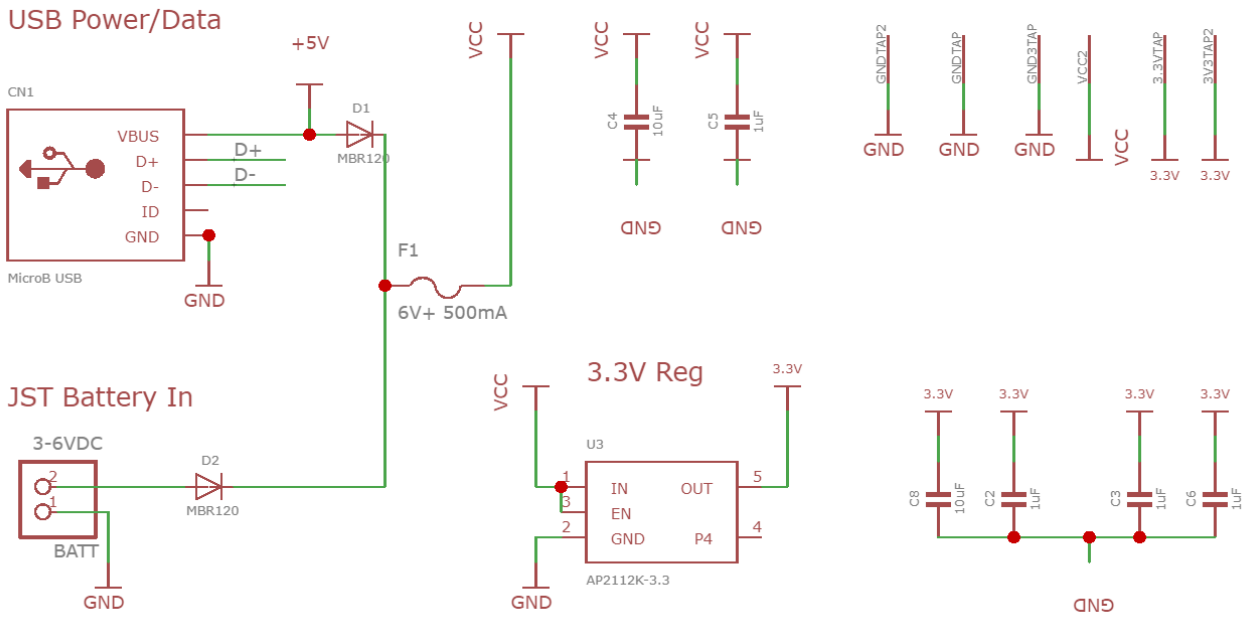
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The Schematic:

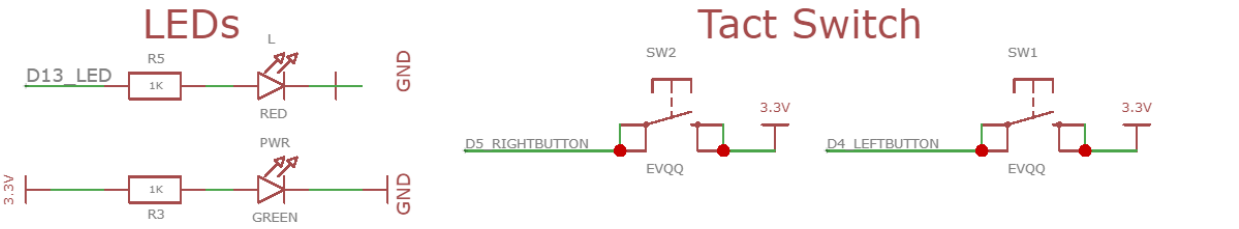
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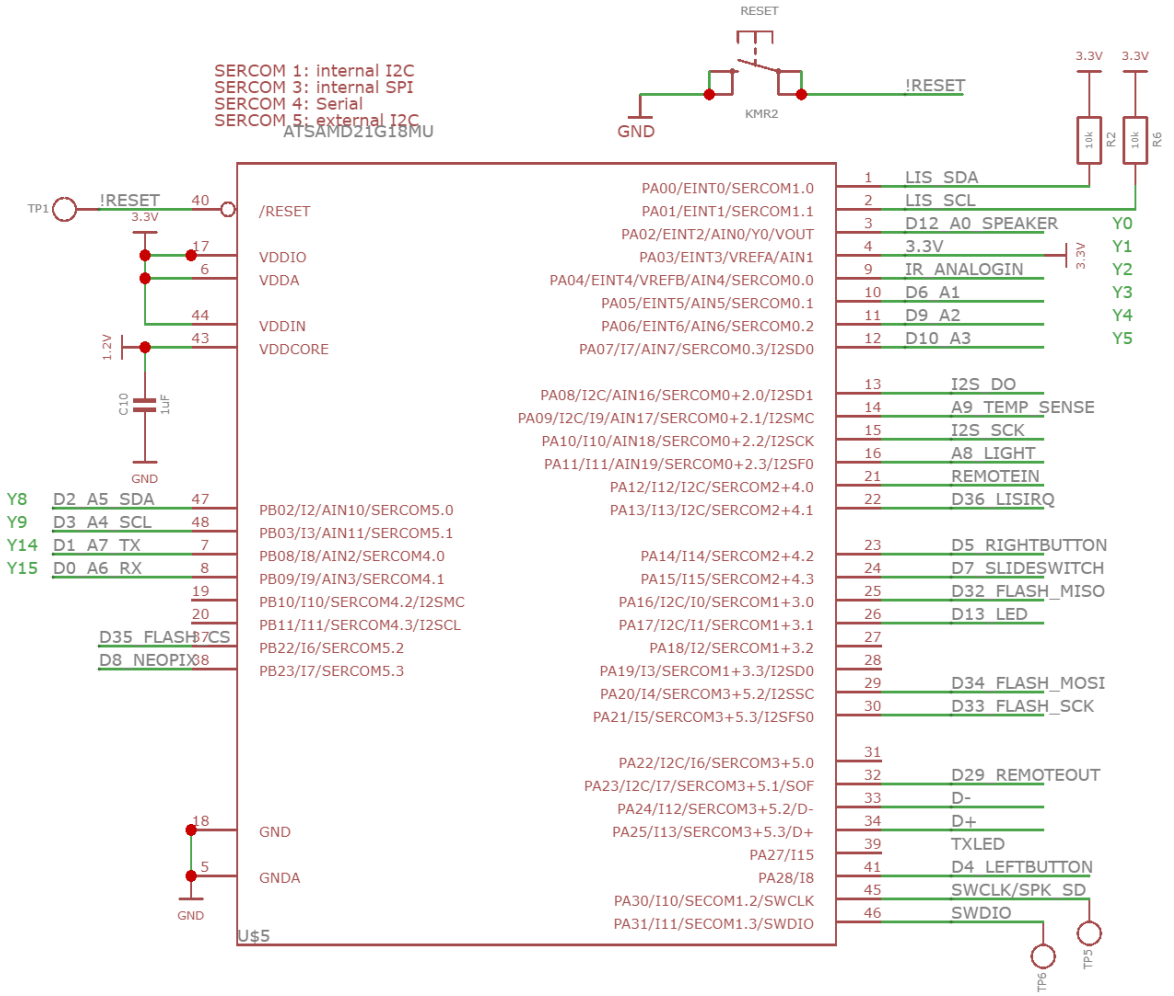
POWER SUPPLY



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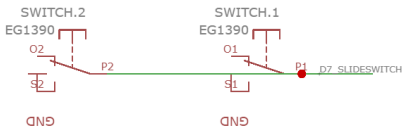


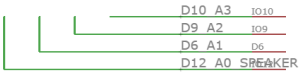
ATSAMD21 MICROCONTROLLER

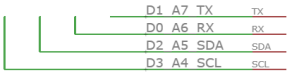


Capacitive Touch/IO Pads

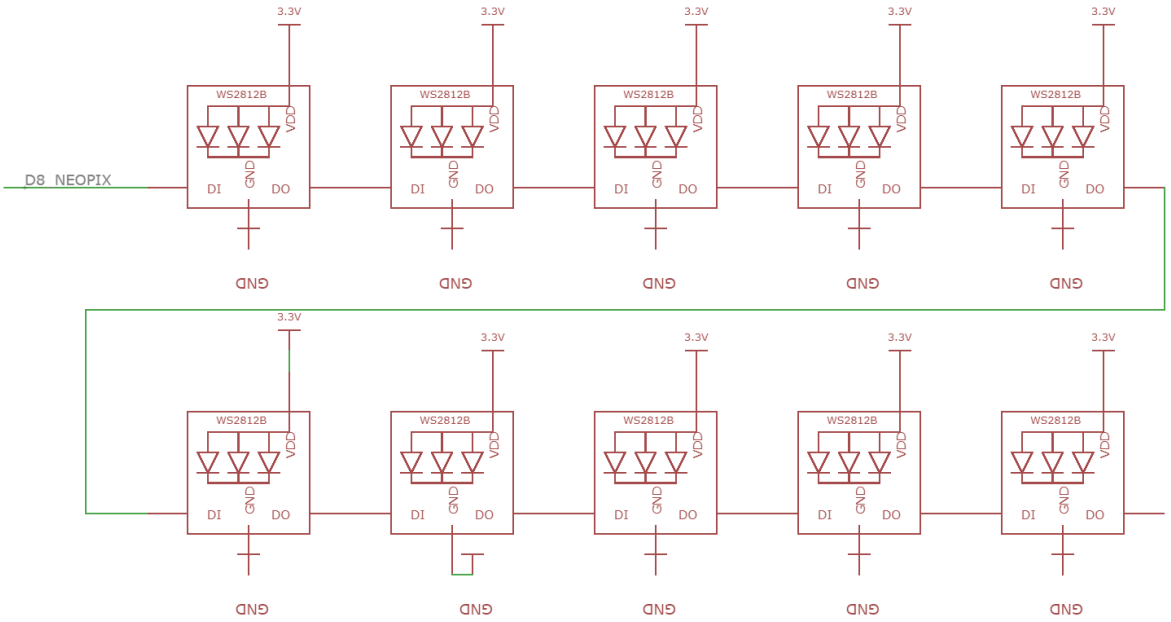
Slide Switch



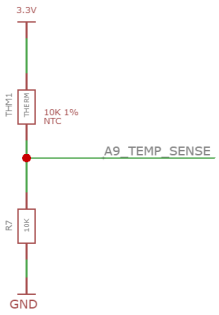




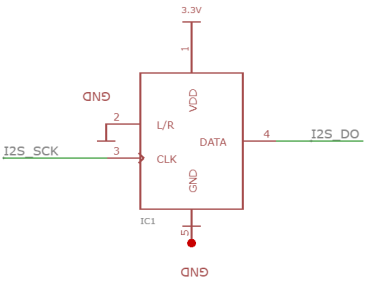
NeoPixels



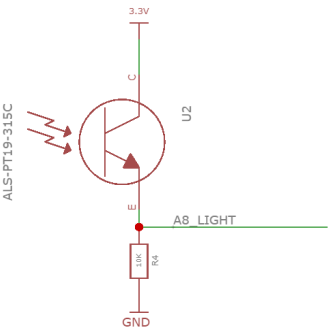
Temp Sensor



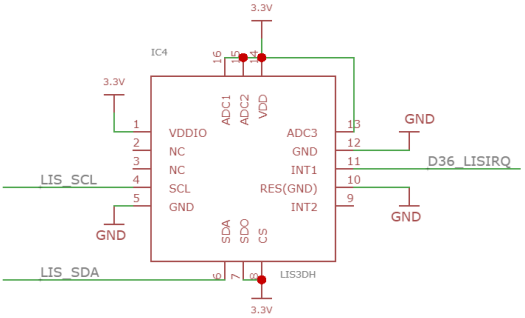
Sound Sensor



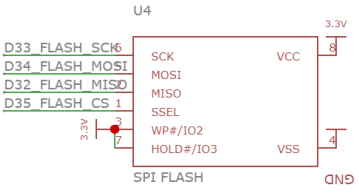
Light Sensor



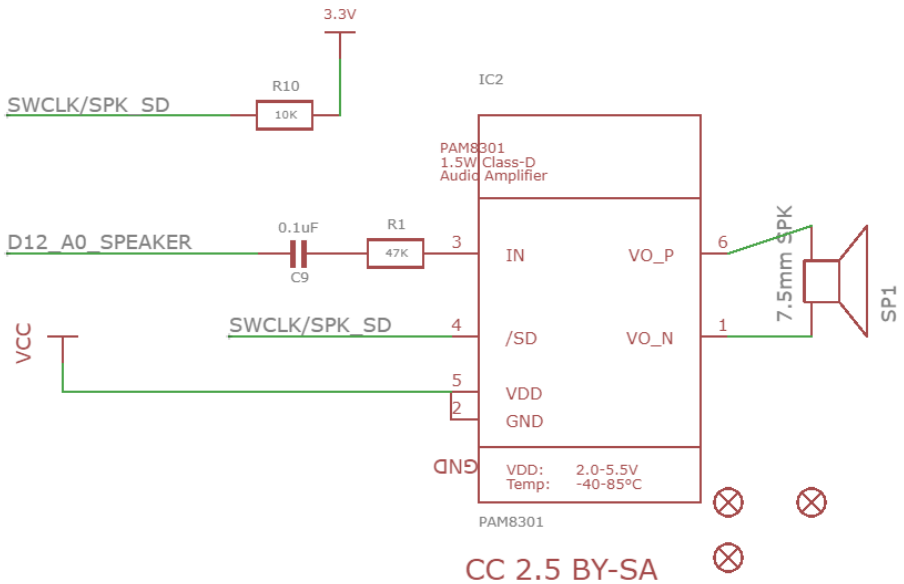
Accelerometer



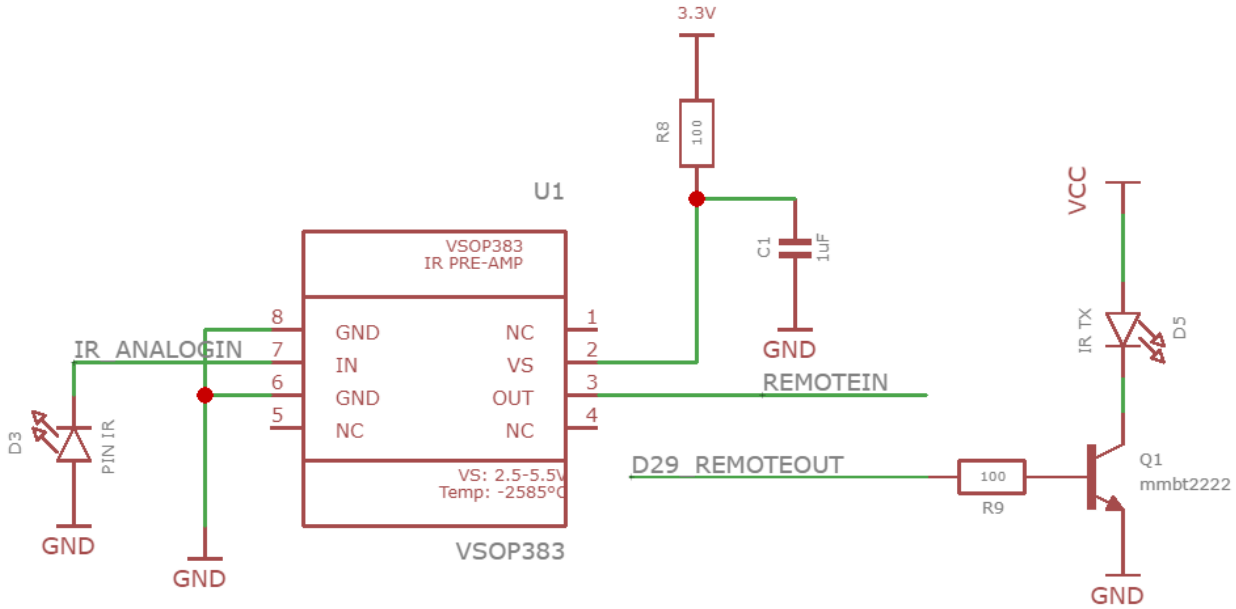
SPI FLASH



Speaker



CC 2.5 BY-SA



Component List:

No.	Component	Designation
1	32-bit ARM Cortex-M0+ Microcontroller ATSAMD21G18MU	U$5
2	2MB Serial NOR Flash Memory GD25Q16C / S25FL116K	U4
3	3-Axis Accelerometer Motion Sensor LIS3DH	IC4
4	1.5W Filterless Class-D Audio Amplifier PAM8301	IC2
5	IR Receiver / Pre-Amplifier IC VSOP383	U1
6	3.3V Low-Dropout Linear Voltage Regulator AP2112K-3.3	U3
7	NPN General Purpose Transistor MMBT2222	Q1
8	Schottky Barrier Rectifier Diode MBR120	D1, D2
9	Ambient Light Sensor / Phototransistor ALS-PT19-315C	U2
10	I2S Digital MEMS Microphone SPH0645LM4H-B	IC1
11	Smart RGB Integrated Light Source WS2812B (5050 RGB LED with integrated IC)	NeoPixel Chain (10 Units)
12	10K 1% NTC Thermistor	THM1
13	Surface-Mount Mini Speaker (7.5mm / 8Ω)	SP1
14	Infrared LED (IR Transmitter)	D5
15	PIN Photodiode (IR Receiver Element)	D3
16	Resettable Fuse 6V+ 500mA	F1
17	Micro-B USB Connector	CN1
18	JST-PH 2-Pin Battery Connector	BATT
19	Reset Switch KMR2	RESET
20	SPDT Slide Switch EG1390	SWITCH.1, SWITCH.2
21	Tactile Push Buttons EVQQ	SW1, SW2
22	Status LEDs (Red "L", Green "PWR")	L, PWR
23	Resistor	R1-R10
24	Capacitor	C1-C10

Datasheets:

- ATSAMD21G18
- GigaDevice GD25Q16C
- LIS3DH
- PAM8301
- VSOP38338
- AP2112
- MMBT2222
- MBR120
- ALS-PT19-315C
- SPK0415HM4H
- WS2812B - Adafruit
- NTC Thermistors
- KMR2 Series
- SERIES EG1390 SWITCHES

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Note:

If EasyEDA doesn't have any footprint of any particular component, you may make or download a suitable footprint from the internet. In that case, follow the dimensions given in the datasheets.

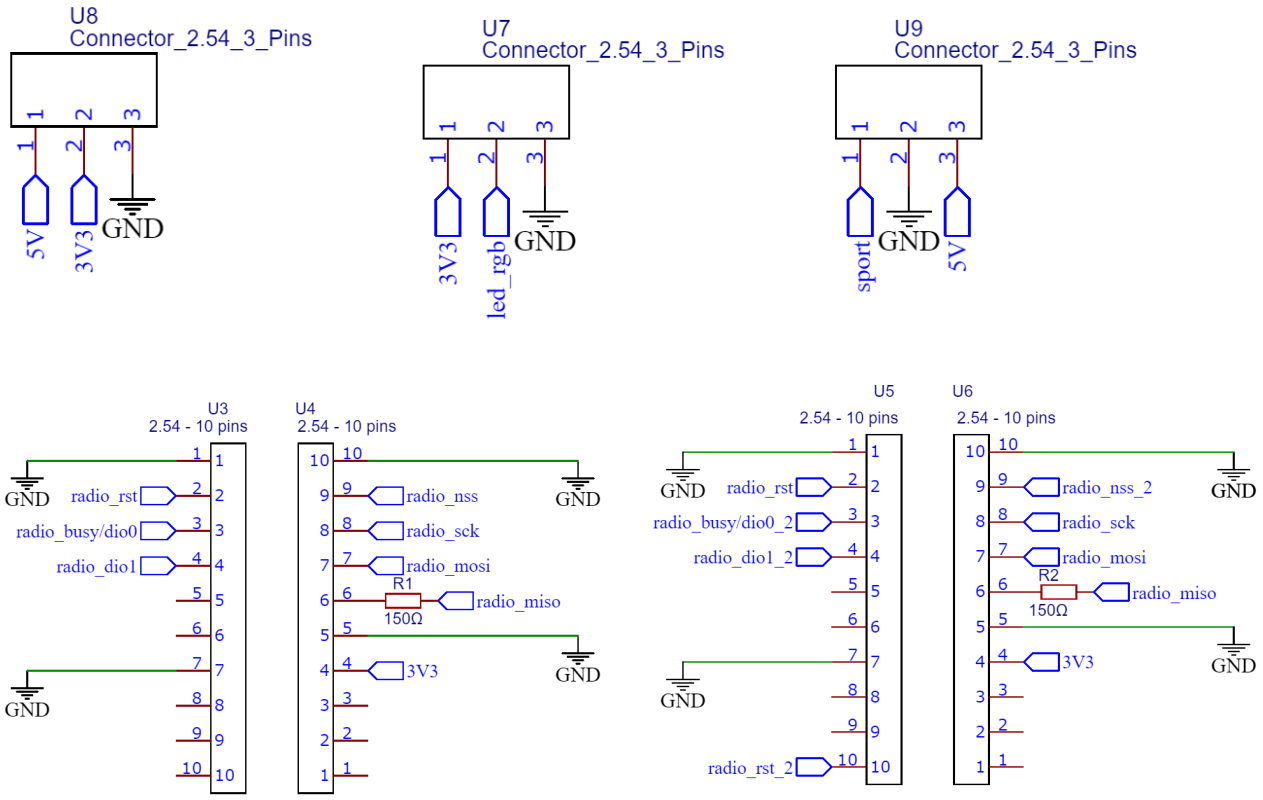
Problem 3: ExpressLRS ESP32 RF Development PCB (70 points)

Design an ESP32-based RF development/control PCB based on the provided editable EasyEDA source. The design should integrate the processing section, power circuitry, programming/debug connections, and RF/control interfaces while applying RF-aware placement and grounding practices.

Design Requirements

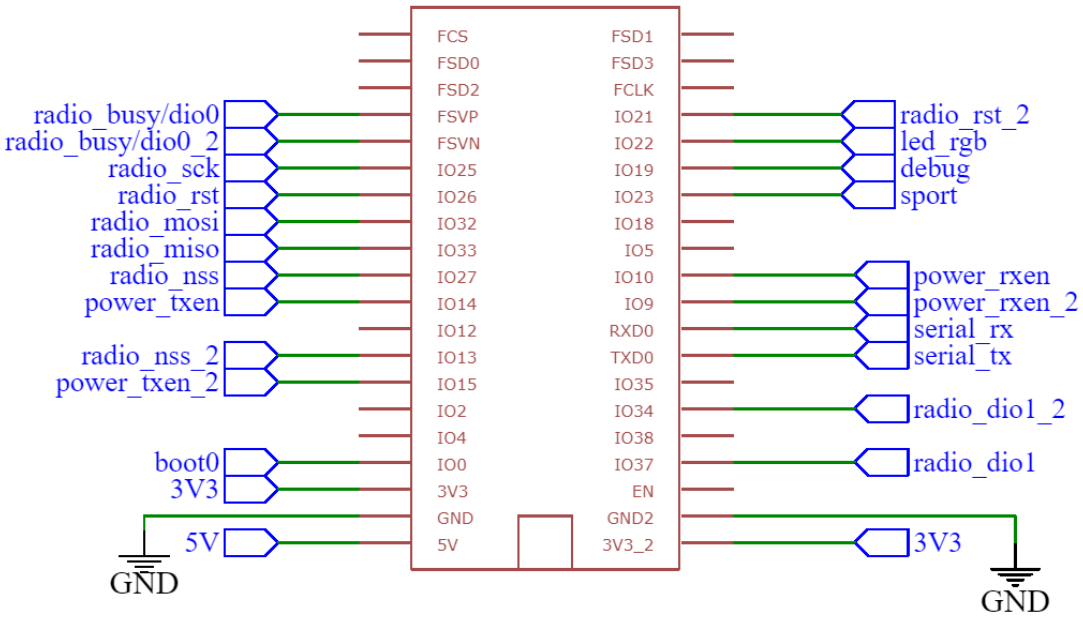
- Keep RF-related components and RF signal paths compact.
- Avoid routing noisy power or high-current traces through the RF section.
- Use a continuous ground reference wherever appropriate around RF/digital sections.
- Place supply decoupling capacitors as close as practical to IC supply pins.
- Keep programming and debug connections readily accessible.
- Place major external connectors along the PCB perimeter.
- Maintain clear antenna/RF keep-out areas where required by the selected RF implementation.
- Use clear net naming, test points, and silkscreen labels.

The Schematic:

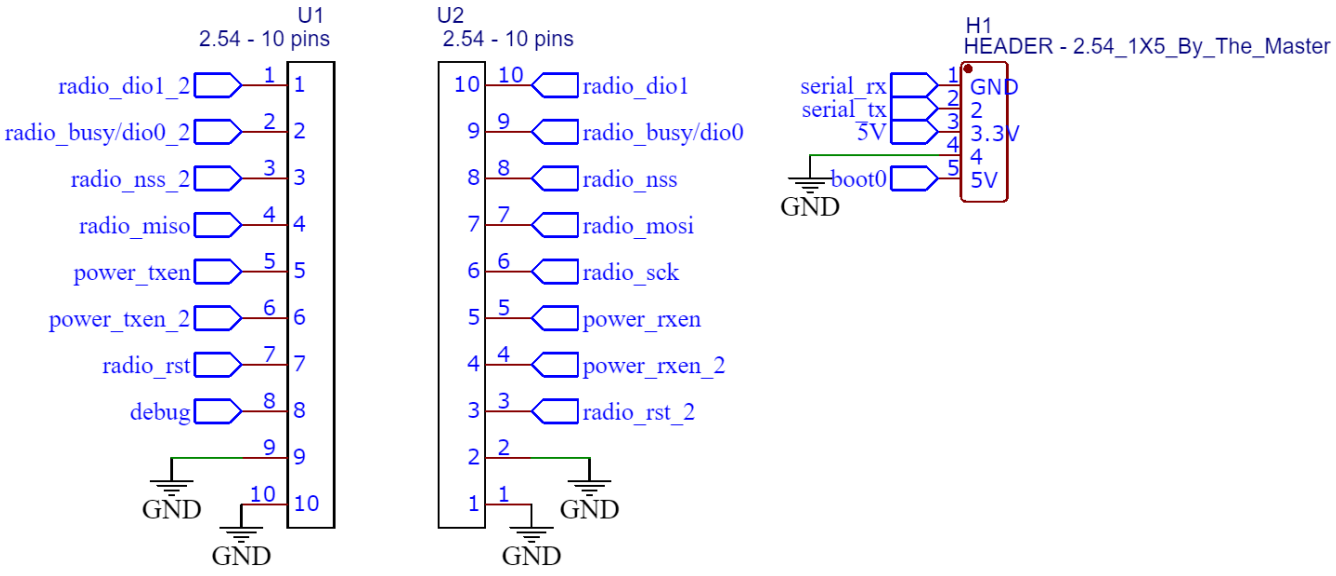


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ESP32PICOKIT_ESP32-PICO-KITPICO



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Component List:

No.	Component	Designation
1	ESP32 Development Board ESP32-PICO-KIT	1
2	1x5 Pin Female Header (2.54mm Pitch)	H1
3	3-Pin Connector (2.54mm Pitch)	U7, U8, U9
4	20-Pin Microcontroller Header (2.54mm Pitch)	U10, U11
5	10-Pin Header / Connector (2.54mm Pitch)	U1-U6
6	150Ω Resistor (0402 Package)	R1, R2

Datasheets:

- ESP32-PICO Series

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Note:

If your designing software doesn't have any footprint of any particular component, you may make or download a suitable footprint from the internet. In that case, follow the dimensions given in the datasheets.

Design and DRC Rules

Follow the rules given in the competition Rulebook.

NOTE

- Participants can add any labels or new test points to the schematic.
- Participants will have to choose a suitable footprint if there is no datasheet provided for any component. If datasheet is provided, participants can choose any footprint that suits the design from the datasheet

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